

Treat the following questions:

Exercise 1: (6 pts)

Elaborate a use case diagram for the following system:

- An employment company has developed a Web site for its services.
- A person looking for a job must register (sign up) on the site after filling a form that contains his name, his date of birth, his nationality, his marital status, and other information. The registration code will be used later, by this person, to « log in » and display or modify his information.
- When a company wants to subscribe to the site, it must sign up first. After that, it can put job offers for employees with certain characteristics. The registered company can also display the persons' applications but cannot modify them.
- The site administrator can display the persons' applications and the companies' job offers as well.
- If a company finds an appropriate person for the requested job, a contract is prepared by the administrator. After that, the contract must be confirmed by both the company and the person by sending each one an email to the administrator. Finally, the person's application and the company's job offer will be removed from the system.

Exercise 2: (5 pts)

Elaborate a class diagram that corresponds to the following specifications. To simplify the exercise, we limit the diagram to one year study.

A university wants to manage the courses available at many colleges. For that, we have the following information:

- Each college has a Web site.
- Each college is structured in departments that have specific teachers. One of the teachers is responsible for the department.
- A teacher is defined by his last name, first name, tel, email, hiring date, and his index.
- A teacher cannot teach more than one course.
- The students may follow many courses and receive one grade for each course.
- For each student, we want to manage his last name, first name, tel, email, as well as his entrance year to the college.
- A course may be taught by many teachers, but it's always given in the same lecturing room. Each room has a determined number of places.
- We should be able to calculate the average per course and per department.
- We should also be able to calculate the general average of a student and display the courses that he has no grades yet.
- Finally, we should be able to print out a specific report that contains the first name, the last name, the tel and the email of a teacher or a student.

Exercise 3: (4 pts)

The normal use of a supermarket fund is as follows:

- The customer arrives to the fund with his articles to pay.
- The cashier registers the identification number of each article as well as the quantity if it's more than 1.
- The cashier displays the price and the label of each item.
- When all the bought articles are registered, the cashier signals the end of the sale.
- The cashier displays the total amount to pay.
- The cashier announces to the customer the total amount to pay.
- The customer chooses his payment mode:
 - Cash: The cashier cashes money, then the fund indicates the amount that should be returned to the customer.
 - Check: The cashier writes the identity number of the customer.
 - Credit card: The authorization demand is sent before the input.
- The fund registers the sale and prints it out.
- The cashier gives the fund's ticket to the customer.

a- Construct the sequence diagram such that the payment mode is cash.

b- Construct the sequence diagram such that the payment mode is credit card.

Exercise 4: (3 pts)

Construct a statechart diagram that represents the following situation:

Consider a car with automatic transmission (automatic gearbox). At start-up, the gearbox is at the neutral position (N). The reverse (R) as well as the park (P) positions can be directly triggered (i.e. used) from N. The first forward march (L1) can also be directly triggered from N. However, the other forward marches (the second and the third) can be triggered in sequence: 1→2→3 for acceleration and 3→2→1 for deceleration. Only R, P, and L1 can be directly returned to N.

Exercise 5: (2 pts)

Describe the functioning of an automatic teller machine (ATM) using an activity diagram as follows:

The customer inserts his card. The validity is automatically verified. Then, he is asked to enter the card's secret code. After three failed tentatives, the card is swallowed (i.e. taken by the ATM machine). Otherwise, the customer can indicate the amount he wishes to withdraw. The balance of his bank account is then checked to make sure that the withdrawal is possible. If the balance is insufficient, the customer will be informed and can therefore enter another amount less than the previous one. If the balance is sufficient, the ATM machine returns the card and delivers the cash money with a receipt.
